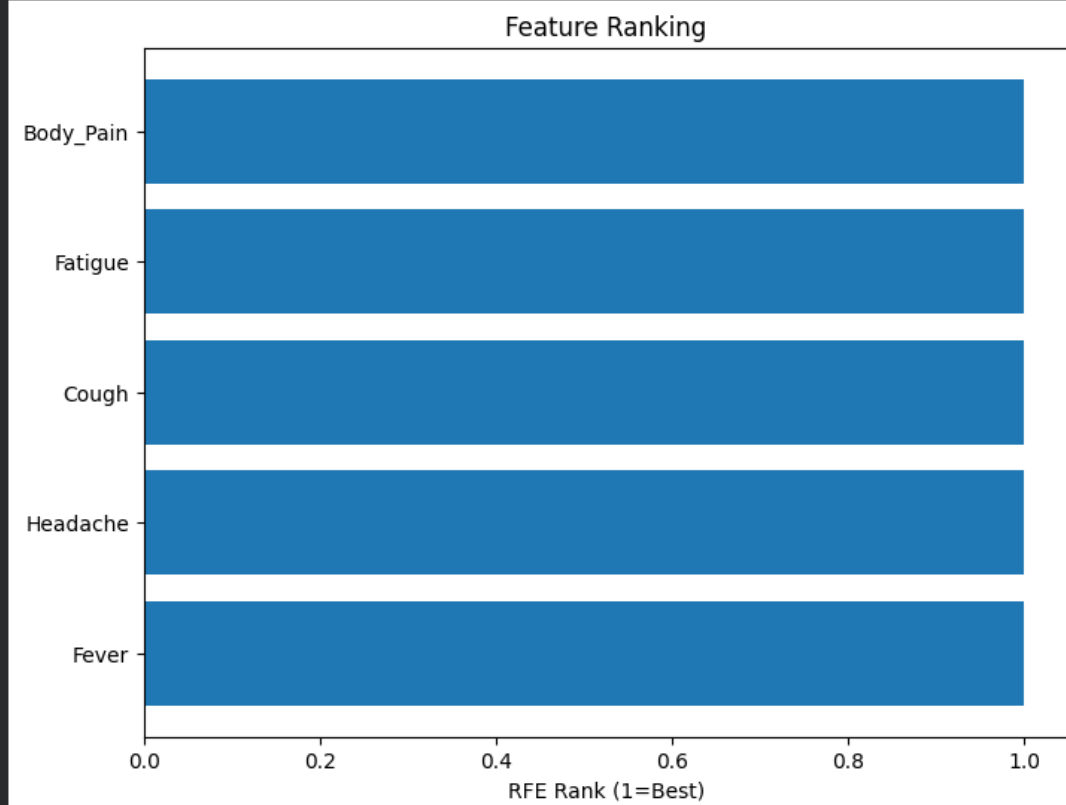


OUTPUTS

```
Dataset shape: (5000, 6)
Columns: Index(['Fever', 'Headache', 'Cough', 'Fatigue', 'Body_Pain', 'Disease'], dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Fever       5000 non-null   float64
1   Headache    5000 non-null   float64
2   Cough       5000 non-null   float64
3   Fatigue     5000 non-null   float64
4   Body_Pain   5000 non-null   float64
5   Disease     5000 non-null   object
dtypes: float64(5), object(1)
memory usage: 234.5+ KB
None
Fever      0
Headache   0
Cough      0
Fatigue    0
Body_Pain  0
Disease    0
dtype: int64
      Fever  Headache    Cough  Fatigue  Body_Pain  Disease
0  98.283156  4.118632  2.687983  4.116661  8.470956  Body Ache
1  98.284660  6.963971  1.960734  5.495257  8.033919  Body Ache
2  103.622552  8.702915  2.854463  6.546993  2.205383    Malaria
3  98.254732  1.354851  5.880676  5.418289  2.695560  Common Cold
4  97.182670  0.357962  6.206723  4.488365  1.406323    Cough
```

```
Top 5 Selected Features: Index(['Fever', 'Headache', 'Cough', 'Fatigue', 'Body_Pain'], dtype='object')
```



Performance with Selected Features:

Accuracy: 0.878

Precision: 0.8790406838029243

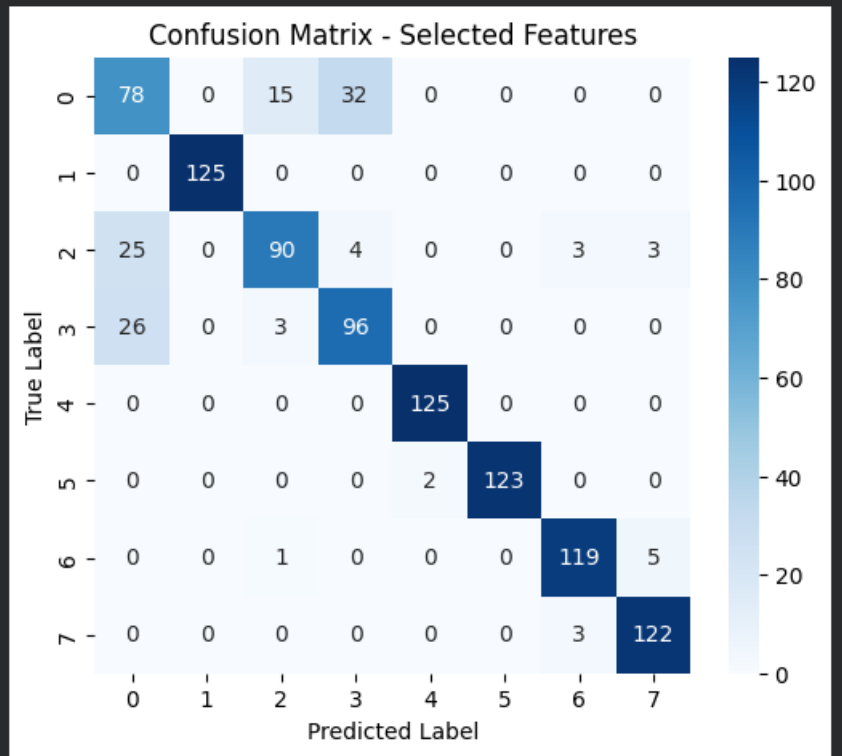
Recall: 0.878

F1 Score: 0.8779184288339977

ROC-AUC (ovr): 0.9858628571428572

Classification Report:

	precision	recall	f1-score	support
0	0.60	0.62	0.61	125
1	1.00	1.00	1.00	125
2	0.83	0.72	0.77	125
3	0.73	0.77	0.75	125
4	0.98	1.00	0.99	125
5	1.00	0.98	0.99	125
6	0.95	0.95	0.95	125
7	0.94	0.98	0.96	125
accuracy			0.88	1000
macro avg	0.88	0.88	0.88	1000
weighted avg	0.88	0.88	0.88	1000



Performance with All Features:
Accuracy: 0.878
Precision: 0.8790406838029243
Recall: 0.878
F1 Score: 0.8779184288339977
ROC-AUC (ovr): 0.9858628571428572